This is one of my two preparation documents for my oral exam, focused on geometric/combinatorial group theory.

Preliminaries

Group-Theoretic Constructions

There are two major ways we may construct new groups out of old: extensions/semidirect products and pushouts/amalgamated free products. We will provide a brief overview of both of these.

Definition: Let Q and N be groups. We say a group G is an extension of Q and N if there is an exact sequence

$$1 \to N \to G \to Q \to 1.$$

If the extension splits — i.e., there is a section $Q \to N$ — then $G \cong Q \ltimes_{\varphi} N$ is a semidirect product for some $\varphi\colon Q \to \mathrm{Aut}(N)$.

Example:

- If the homomorphism $\varphi\colon Q \to \mathrm{Aut}(N)$ is trivial, then $N \rtimes_{\varphi} Q \cong N \times Q$ is a direct product.
- The map $\mathbb{Z}/2\mathbb{Z} \to \mathrm{Aut}(\mathbb{Z}/n\mathbb{Z})$ given by inversion yields the dihedral group

$$D_n \cong \mathbb{Z}/2\mathbb{Z} \ltimes \mathbb{Z}/n\mathbb{Z}.$$

- If G is any group, the lamplighter group of G is the semidirect product $\mathbb{Z} \ltimes G^{\mathbb{Z}}$ with homomorphism $\varphi\colon \mathbb{Z} \to \mathrm{Aut}\left(G^{\mathbb{Z}}\right)$ given by

$$\varphi(z) = ((g_n)_{n\in\mathbb{Z}} \mapsto (g_{n+z})_{n\in\mathbb{Z}}).$$

- the wreath product of G and H, denoted $G \wr H$, is the semidirect product $H \ltimes G^H$ with $\varphi\colon H \to \mathrm{Aut}\left(G^H\right)$ given by the shift action

$$\varphi(h) = \left((g_k)_{k\in H} \mapsto (g_{kh})_{k\in H}\right).$$

Definition: Let A be a group, $\alpha_1 : A \to G_1$, and $\alpha_2 : A \to G_2$ group homomorphisms. We say a group G is a *pushout* of G_1 and G_2 over A with respect to α_1 and α_2 if there are homomorphisms $\beta_1 : G_1 \to G$ and $\beta_2 : G_2 \to G$ satisfying $\beta_1 \circ \alpha_1 = \beta_2 \circ \alpha_2$ and the universal property for pushouts.

That is, for every group H and all group homomorphisms $\varphi_1 : G_1 \to H$ and $\varphi_2 : G_2 \to H$ satisfying $\varphi_1 \circ \alpha_1 = \varphi_2 \circ \alpha_2$, there is exactly one homomorphism $\varphi\colon G \to H$ satisfying $\varphi \circ \beta_i = \varphi_i$ for each $i = 1, 2$. We denote the pushout by $G_1 *_A G_2$.

- If $A = \{e\}$ is the trivial group, then the pushout $G_1 *_A G_2 = G_1 * G_2$ is called the *free product* of G_1 and G_2.
- If α_1 and α_2 are injective, then $G_1 *_A G_2$ is called an amalgamated free product of G_1 and G_2 over A.

Example:

- The free group F_n on n letters is the n-fold free product $\mathbb{Z} * \cdots * \mathbb{Z}$.
- The infinite dihedral group $D_\infty \cong \mathbb{Z}/2\mathbb{Z} \ltimes \mathbb{Z}$ is isomorphic to the free product $\mathbb{Z}/2\mathbb{Z} * \mathbb{Z}/2\mathbb{Z}$.
- The matrix group $\mathrm{SL}_2(\mathbb{Z})$ is isomorphic to the amalgamated free product $\mathbb{Z}/6\mathbb{Z} *_{\mathbb{Z}/2\mathbb{Z}} \mathbb{Z}/4\mathbb{Z}$.
- If X and Y are pointed, non-empty, path-connected topological spaces with nonempty and path-connected intersection, then the fundamental group of their union (which is a type of pushout) is given by the amalgamated free product of their fundamental groups over the fundamental group of the intersection.

Theorem: Pushout groups exist and are unique up to isomorphism.

Proof. Take presentations

$$G_i = \left\langle \left\{ x_g : g \in G_i \right\} \middle| R_i \right\rangle.$$

Then, the presentation

$$G := \left\langle \left\{ x_g : g \in G_1 \right\} \cup \left\{ x_g : g \in G_2 \right\} \middle| \left\{ x_{\alpha_1(a)} x_{\alpha_2(a)}^{-1} : a \in A \right\} \cup R_1 \cup R_2 \right\rangle.$$

The group homomorphisms $\beta_i \colon G_i \to G$ taking $g \mapsto x_g$, along with the relations $\left\{ x_{\alpha_1(a)} x_{\alpha_2(a)}^{-1} : a \in A \right\}$, give $\beta_1 \circ \alpha_1 = \beta_2 \circ \alpha_2$. The universal property then follows from the universal property of generators and relations. □

Definition: Let G be a group, and let $A, B \subseteq G$ be isomorphic subgroups with isomorphism $\theta \colon A \to B$. The *HNN extension* of G with respect to θ is the group

$$G*_\theta := \left\langle \left\{ x_g : g \in G \right\} \cup \{t\} \middle| \left\{ t^{-1} x_a t x_{\theta(a)}^{-1} : a \in A \right\} \cup R \right\rangle,$$

where R is the relations for G. We say that t is the stable letter of the HNN extension.

The HNN extension is the universal (in a sense) way to force isomorphic subgroups to be conjugate to each other.

Group Homology/Cohomology

Definition: Let R be a commutative unital ring. The *group ring* is the ring $R[G]$ with

- underlying abelian group $\bigoplus_{g \in G} Rg$, where the g are formal symbols corresponding to the elements of G;
- multiplication given by the R-bilinear extension of the group multiplication $(g, h) \mapsto gh$.

If G is a non-abelian group and R is non-trivial, then the group ring $R[G]$ is not commutative.

Definition: Let G be a group.

- The *simplicial resolution* of G, denoted $C_*(G)$, is the chain complex of $\mathbb{Z}$-modules whose chains are defined by

  $$C_n := \bigoplus_{G^{n+1}} \mathbb{Z},$$

  with boundary maps given by $\partial_n \colon C_n(G) \to C_{n-1}(G)$ taking

  $$\partial_n \left(\sum_{g \in G^{n+1}} a_g (g_0, \dots, g_n) \right) = \sum_{g \in G^n} a_g \sum_{j=0}^{n} (-1)^j \left(g_0, \dots, g_{j-1}, g_{j+1}, \dots, g_n \right).$$

- If A is a left $\mathbb{Z}[G]$-module, then the simplicial chain complex of G with A-coefficients is given by $C_n(G; A) = A \otimes_{\mathbb{Z}[G]} C_*(G)$ with boundary operator $\partial_*^A = \mathrm{id}_A \otimes_{\mathbb{Z}[G]} \partial_*$. The elements of $\ker\left(\partial_n^A\right)$ are called the n-cycles.

  The n-th homology of G with A-coefficients is the $\mathbb{Z}$-module given by

  $$H_n(G; A) := \frac{\ker\left(\partial_n^A \colon C_n(G;A) \to C_{n-1}(G;A)\right)}{\operatorname{im}\left(\partial_{n+1}^A \colon C_{n+1}(G;A) \to C_n(G;A)\right)}.$$

- Let A be a left $\mathbb{Z}[G]$-module. The simplicial cochain complex of G with A-coefficients is defined by

  $$C^*(G; A) := \mathrm{Hom}_{\mathbb{Z}[G]}(C_*(G), A)$$

with coboundary operator ∂_A^* given by the A-dual of ∂_*. Elements of $\ker\left(\delta_A^n\right)$ are called the n-cocycles. The nth cohomology of G with A-coefficients is the $\mathbb{Z}$-module

$$H^n(G;A) := \frac{\ker\left(\partial_A^n \colon C^n(G;A) \to C^{n+1}(G;A)\right)}{\operatorname{im}\left(\partial_A^{n-1} \colon C^{n-1}(G;A) \to C^n(G;A)\right)}.$$

Nilpotent Groups

Lemma: Let G be a group, let $N \trianglelefteq G$ be a normal subgroup, and let $H, K, L \leq G$ be subgroups. If $[[H,K],L]$ and $[[K,L],H]$ are contained in N, then so is $[[L,H],K]$.

Proof. The subgroups $[[H,K],L]$, $[[K,L],H]$, and $[[L,H],K]$ are generated by the commutators $[h,k,\ell]$, $[k,\ell,h]$, and $[\ell,h,k]$ respectively, where $h \in H$, $k \in K$, and $\ell \in L$. The Hall–Witt identity says that

$$\begin{aligned} e &= b^{-1}\left[a, b^{-1}, c\right]b \\ &= c^{-1}\left[b, c^{-1}, a\right]c \\ &= a^{-1}\left[c, a^{-1}b\right]a, \end{aligned}$$

so all of these commutators are contained in N. □

Definition: Let G be a group. The *lower central series* of G is the sequence $(\gamma_k(G))_{k\geq 1}$ of subgroups defined recursively by

$$\begin{aligned} \gamma_1(G) &:= G \\ \gamma_{k+1}(G) &:= [\gamma_k(G), G]. \end{aligned}$$

We say a group G is *nilpotent* if its lower central series is finite. That is, if there is some integer $k \geq 1$ such that $\gamma_k(G) = e$. The smallest such k is called the *nilpotence class* (or nilpotency class) of G.

Inductively, we observe that $\gamma_k(G)$ is characteristic in G with $\gamma_{k+1}(G) \subseteq \gamma_k(G)$, meaning that the lower central series is a descending normal series.

Lemma: Let G be a group. Then,

$$\left[\gamma_i(G), \gamma_j(G)\right] \subseteq \gamma_{i+j}(G).$$

Proof. We prove by induction on i. By the definition of γ_{i+1}, we have that $\left[\gamma_1(G), \gamma_j(G)\right] = \left[G, \gamma_j(G)\right] = \gamma_{j+1}(G)$.

Now, we show that

$$\left[\gamma_{i+1}(G), \gamma_j(G)\right] \subseteq \gamma_{i+j+1}(G).$$

First, we claim that given $a, b \in \gamma_{i+1}(G)$ and $c \in \gamma_j(G)$, we have

$$\begin{aligned} [ab, c] &= [a,c][b,c] \bmod \gamma_{i+j+1}(G) \\ \left[a^{-1}, c\right] &= [a,c]^{-1} \bmod \gamma_{i+j+1}(G). \end{aligned}$$

Since $a \in \gamma_{i+1}(G) \subseteq \gamma_i(G)$, the induction hypothesis has $[a,c] \in \left[\gamma_i(G), \gamma_j(G)\right] \subseteq \left[\gamma_{i+j}(G)\right]$. Consequently, $[[a,c],b] \in \left[\gamma_{i+j}(G), \gamma_j(G)\right] \subseteq \left[\gamma_{i+j}(G), G\right] = \gamma_{i+j+1}(G)$.

By standard commutator identities, we have

$$\begin{aligned} [ab, c] &= [a,c][a,c,b][b,c] \\ &= [a,c][b,c]\left([b,c]^{-1}[a,c,b][b,c]\right). \end{aligned}$$

Since $\gamma_{i+j+1}(G)$ is normal, it follows that $[b,c]^{-1}[a,c,b][b,c] \in \gamma_{i+j+1}(G)$, so this gives our first desired result. Similarly, by taking $b = a^{-1}$, we obtain the second desired result.

Therefore, we only need to show that $[a,x,b] \in \gamma_{i+j+1}(G)$ For all $a \in \gamma_i(G)$, $x \in G$, and $b \in \gamma_j(G)$.

From the Hall–Witt identity, we may write

$$[a,x,b] = x^{-1}\left(\left(a^{-1}\left[b,a^{-1}x^{-1}\right]a\right)^{-1}\left(b^{-1}\left[x^{-1},b^{-1}a\right]b\right)^{-1}\right)x.$$

while by the inductive hypothesis, $\left[b,a^{-1},x^{-1}\right] \in \left[\gamma_{i+j}(G),G\right] = \gamma_{i+j+1}(G)$, and $\left[x^{-1},b^{-1},a\right] \in \left[\gamma_{j+1}(G),\gamma_i(G)\right] \subseteq \gamma_{i+j+1}(G)$, seeing as $\gamma_{i+j+1}(G)$ is a normal subgroup (hence the entire conjugacy class must land in it if any conjugate lands in it). □

We say a descending series $G = G_1 \geq G_2 \geq \cdots \geq G_k \geq G_{k+1} \geq \cdots$ is *central* if $\left[G_i,G_j\right] \subseteq G_{i+j}$ for all $i,j \geq 1$. This condition implies that each G_i is normal in G.

Lemma: Let G be a group. Suppose $(G_k)_{k\geq 1}$ is a descending central series for G. Then, $G_i \supseteq \gamma_i(G)$ for each $i \geq 1$.

Proof. We proceed by induction. First, $G_1 = G = \gamma_1(G)$.

Suppose the statement holds for i. The defining property of a central series and the inductive hypothesis give

$$\begin{aligned} G_{i+1} &\supseteq [G_i,G] \\ &\supseteq [\gamma_i(G),G] \\ &= \gamma_{i+1}(G). \end{aligned}$$

□

In particular, this means that a group G is nilpotent if and only if it admits any finite descending central series.

Proposition: Let G be a nilpotent group with nilpotence class c. Then, any subgroup and any quotient of G is nilpotent of nilpotence class at most c.

Proof. If $H \leq G$ is a subgroup, inductively we have $\gamma_i(H) \subseteq \gamma_i(G)$.

If $\varphi\colon G \to \overline{G}$ is a surjective homomorphism, then we have $\varphi([H,K]) = [\varphi(H),\varphi(K)]$ for all subgroups $H,K \leq G$. In particular, this means that inductively we have $\gamma_{i+1}(\varphi(G)) = \varphi(\gamma_{i+1}(G))$. □

Definition: Let G be a group. The *upper central series* of G is the ascending series $(Z_k(G))_{k\geq 0}$, where

$$\begin{aligned} Z_0(G) &= e \\ Z_{i+1}(G) &= \pi_i^{-1}(Z(G/Z_i(G))), \end{aligned}$$

where $\pi_i\colon G \to G/Z_i(G)$ is the projection homomorphism, and $Z(H)$ is the center of a group H.

In particular, we have $Z(G/Z_i(G)) = Z_{i+1}(G)/Z_i(G)$.

Finitely Generated Nilpotent Groups

We start by considering torsion-free finitely generated nilpotent groups. That is, finitely generated nilpotent groups where no nontrivial element has finite order.

Lemma: Let G be a torsion free (not necessarily finitely generated) nilpotent group with upper central series $(Z_k(G))_{k\geq 0}$. Then, all the factors $Z_{i+1}(G)/Z_i(G)$ are torsion-free abelian groups.

Proof. We use induction on nilpotence class. If $c = 1$, then $Z_1(G)/Z_0(G) = G$ is abelian, so it is a torsion-free abelian group by assumption.

Suppose it holds true for all nilpotent groups of nilpotence class $(c-1)$. Let G have nilpotence class c. Set $H = G/Z_1(G)$. Then, we have $Z_i(H) = Z_{i+1}(G)/Z_i(G)$ by induction and the definition of the upper central series. In particular, H has nilpotence class $(c-1)$.

By the third isomorphism theorem, we have $Z_{i+1}(G)/Z_i(G) \cong (Z_{i+1}(G)/Z_1(G))/(Z_i(G)/Z_1(G)) \cong Z_i(H)/Z_{i-1}(H)$, so it suffices to show that H is torsion-free to complete the proof.

Let $a \in G$. Suppose there exists an integer $m \geq 1$ such that $a^m \in Z_1(G)$, and suppose toward contradiction that $a \notin Z_1(G)$. Notice that $Z_1(G) = Z(G)$, so by assumption there is some $b \in G$ such that $[a,b] \neq e$. Let $1 \leq i \leq c-1$ be the maximal index for which there exists some $b \in \gamma_i(G)$ such that $[a,b] \neq e$, where $(\gamma_i(G))_{i=1}^{c+1}$ is the lower central series. By the various commutator identities, we have

$$\begin{aligned} e &= [a^m, b] \\ &= \left[aa^{m-1}, b\right] \\ &= a^{1-m}[a,b]a^{m-1}\left[a^{m-1}, b\right]. \end{aligned}$$

Since $[a,b] \in \gamma_{i+1}(G)$, we must have that a commutes with $[a,b]$, so $a^{1-m}[a,b]a^{m-1} = [a,b]$, and we have $e = [a^m, b] = [a,b]\left[a^{m-1}b\right]$. Iterating this process gives $e = [a,b]^m$, but since G is torsion free, this gives that $[a,b] = e$, which is a contradiction of the definition of b. Thus, $a \in Z_1(G)$, hence $H = G/Z_1(G)$ is torsion-free. □

Groups and Graphs

One of the first connections between groups and geometry is the notion of a Cayley graph, and the way that the action of a group on a graph is connected to the structure of the group.

Graph Theory Review

Definition:

- A graph is a pair $X = (V, E)$, where $E \subseteq V^{[2]} := \{e : e \subseteq V, |e| = 2\}$. Elements of E are denoted by $\{v_1, v_2\}$ if the graph is undirected. If the graph is directed, then elements of E are denoted by tuples (v_1, v_2).
- If (V, E) is a graph, then we say vertices $v, v' \in V$ are adjacent if $\{v, v'\} \in E$. The number of neighbors is the degree of the vertex.
- If $X = (V, E)$ and $X' = (V', E')$ are graphs, then we say X and X' are isomorphic if there is a graph isomorphism between X and X' — that is, a bijection $f: V \to V'$ such that for any $v, w \in V$, we have $\{v, w\} \in E$ if and only if $\{f(v), f(w)\} \in E'$.

Definition: Let $X = (V, E)$ be a graph.

- A path in X is a sequence $[v_0, \dots, v_n]$ of distinct vertices such that $\left\{v_j, v_{j+1}\right\} \in E$.
- A graph is called connected if every pair of its vertices can be connected by a path in X.
- A cycle of length n is a path $[v_0, \dots, v_n]$ such that $\{v_n, v_0\} \in E$.
- A tree is a connected graph without cycles. Note that this is equivalent to a graph where every pair of vertices has exactly one path connecting them.
- If X is a connected graph, a *spanning tree* for X is a tree that contains all the vertices of X.

In algebraic topology, we use spanning trees as a way to compute fundamental groups. This is done by using the fact that spanning trees are CW subcomplexes (viewing the graph as a CW complex).

Cayley Graphs

Definition: Let G be a group, and let $S \subseteq G$ be a generating set for G without the identity. The Cayley graph of G with respect to S is the graph $\mathrm{Cay}(G,S)$, with

- vertices given by elements of G;
- edges given by
$$\left\{\{g,gs\} : g \in G, s \in S \cup S^{-1}\right\}.$$

Example:

- The Cayley graph of $\mathbb{Z}$ with respect to $\{1\}$ appears as an infinite line, while the Cayley graph of $\mathbb{Z}$ with respect to the generating set $\{2,3\}$ appears as a particular 4-regular graph. See [Löh17, Figure 3.3].
- The Cayley graph of $\mathbb{Z}^2 \subseteq \mathbb{R}^2$ with respect to the generating set $\{(1,0),(0,1)\}$ appears as the integer lattice.
- The Cayley graph of $\mathbb{Z}/6\mathbb{Z}$ with respect to $\{[1]\}$ is a cycle graph.
- The Cayley graph of the symmetric group S_3 with respect to the generating set $\{\sigma,\tau\}$, where we have
$$S_3 := \left\langle \sigma,\tau \middle| \sigma^3,\tau^2,\tau\sigma\tau\sigma\right\rangle.$$
- The Cayley graph of any group with respect to the generating set given by the group itself is a complete graph.

One of the seemingly simple questions one might ask is when exactly finitely generated groups G and H with finite generating sets $S \subseteq G$ and $T \subseteq H$ are such that $\mathrm{Cay}(G,S) \cong \mathrm{Cay}(H,T)$ are isomorphic as graphs. This is essentially a question about Cayley graphs whose automorphisms are affine (i.e., given by a group isomorphism and translation by a fixed group element).

- Cayley graphs of finitely generated abelian groups satisfy the rigidity question; the automorphisms are affine on the free part, and Cayley graphs remember the rank of the group and the size of the torsion subgroup.
- Automorphisms of Cayley graphs of finitely generated torsion-free nilpotent groups are affine.
- Finitely generated free groups admit isomorphic Cayley graphs if and only if they have the same rank.

Given a presentation of a group, the presentation complex is a two-dimensional CW complex given by taking one zero-cell, adding circles for every generator, and attaching disks for every relation (attached by the word for the relation).

The presentation complex associated to $\left\langle x,y \middle| xyx^{-1}y^{-1}\right\rangle$. The presentation complex associated to $\left\langle x \middle| x^2\right\rangle$ is $\mathbb{RP}^2$.

In general, any group admits a classifying space (known as the Eilenberg MacLane space) whose fundamental group is the underlying group and whose higher homotopy groups are trivial. These spaces are unique up to homotopy equivalence. For example, the torus is a classifying space for $\mathbb{Z}^2$ and $\mathbb{RP}^\infty$ is a classifying space for $\mathbb{Z}/2\mathbb{Z}$.

The 1-skeleton of the universal cover of the presentation complex of the presentation $\langle X|R\rangle$ is basically the Cayley graph (save for generators of order 2).

Theorem: Let F be a free group freely generated by $S \subseteq F$. Then, $\mathrm{Cay}(F,S)$ is a tree.

Proof. Every element of F is a unique word in elements of S, so there is a unique path between the identity and any word, so there is a unique path between any two elements, so $\mathrm{Cay}(F,S)$ is a tree. □

There is a converse, but it requires a caveat. In particular, we see that $\mathrm{Cay}(\mathbb{Z}/2\mathbb{Z},[1])$ is a one-edge two-vertex graph (which is a tree).

Theorem: Let G be a group, $S \subseteq G$ a generating set satisfying $st \neq e$ for all $s, t \in S$. If $\mathrm{Cay}(G, S)$ is a tree, then $G \cong F(S)$ is freely generated by S.

Proof. Let S and G be such that $\mathrm{Cay}(G, S)$ is a tree. We will show that G is isomorphic to $F(S)$ by an isomorphism that is the identity on S.

First, we observe that, by the universal property applied to the inclusion $S \subseteq G$, there is a group homomorphism $\varphi\colon F(S) \to G$ that is the identity on S. Since S generates G, it follows that φ is surjective.

Assume toward contradiction that φ is not injective. Let $s_1 \cdots s_n \in F(S) \setminus \{\varepsilon\}$ be a word of minimal length mapped to $e \in G$ by φ.

First, since $\varphi|_S = \mathrm{id}_S$, we must have $n > 1$.

Next, if $n = 2$, then we would have

$$\begin{aligned} e &= \varphi(s_1 s_2) \\ &= \varphi(s_1)\varphi(s_2) \\ &= s_1 s_2, \end{aligned}$$

contradicting the fact that $s_1 \cdots s_n$ is reduced with $st \neq e$ in G.

Next, suppose $n \geq 3$. Inductively, we may consider the sequence of elements $g_0, \dots, g_{n-1}$ in G given by $g_{j+1} = g_j s_{j+1}$ for all $0 \leq j \leq n-2$. By minimality of the word $s_1 \cdots s_n$, the elements $g_0, \dots, g_{n-1}$ are distinct, and since $g_n = s_1 \cdots s_n = e$, it follows that $[g_0, \dots, g_{n-1}]$ is a cycle in $\mathrm{Cay}(G, S)$. Yet, this contradicts the fact that $\mathrm{Cay}(G, S)$ is a tree. Thus, φ is injective. ☐

Free Groups and Actions on Trees

Quasi-Isometry of Metric Spaces and Groups

Growth types of Groups

Definition: Let G be a finitely generated group with symmetric generating set S. The *word length* of $g \in G$ with respect to S, denoted $\ell_S(g)$, is the smallest $n \geq 0$ such that g may be written as a product of n elements of S. That is,

$$\ell_S(g) = \min\{n : g = s_1 s_2 \cdots s_n \text{ with } s_i \in S \text{ and } n \geq 0\}.$$

The word metric is then

$$d_S(g, h) = \ell_S\left(g^{-1} h\right).$$

Proposition: The word metric is a left-invariant metric. That is, it satisfies the following:

(i) $d_S(g, h) = 0$ if and only if $g = h$;

(ii) $d_S(g, h) = d_S(h, g)$;

(iii) $d_S(g, h) \leq d_S(g, k) + d_S(k, h)$;

(iv) $d_S(g, h) = d_S(kg, kh)$.

Proof.

(i) We have that $\ell(g) = 0$ if and only if $g = e$ by the fact that any nontrivial element must be a nontrivial product of elements of S.

(ii) Notice that $\ell(g) = \ell\left(g^{-1}\right)$ since S is symmetric, so if

$$g = s_1 \cdots s_n$$

realizes $\ell(g)$, then

$$g^{-1} = s_n^{-1} \cdots s_1^{-1}$$

must realize $\ell\left(g^{-1}\right)$, as otherwise we would have a smaller representation for g in terms of a product of elements of S.

(iii) Observe that, by the existence of relations and word reductions, we have $\ell(gh) \leq \ell(g) + \ell(h)$.

(iv) We have

$$\begin{aligned} d(kg, kh) &= \ell\left((kg)^{-1}(kh)\right) \\ &= \ell\left(g^{-1}k^{-1}kh\right) \\ &= \ell\left(g^{-1}h\right) \\ &= d(g,h). \end{aligned}$$

□

Amenability of Groups

Residually Finite Groups

Soficity and Hyperlinearity

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